← Retour au choix des chapitres

Correction - Calculs sur les fractions

Exercice n°1

a. $(\frac{5}{4} - \frac{7}{2})(\frac{2}{3} + \frac{1}{6})$

$= (\frac{5}{4} - \frac{7\times2}{2\times2})(\frac{2\times2}{3\times2} + \frac{1}{6})$

$= (\frac{5}{4} - \frac{14}{4})(\frac{4}{6} + \frac{1}{6})$

$= \frac{-9}{4} \times \frac{5}{6}$

$= \frac{-45}{24}$

$= \frac{-\cancel{3}\times3\times5}{2\times2\times2\times\cancel{3}}$

$= \frac{-15}{8}$

b. $(\frac{1}{4} - \frac{1}{5}) \times (7 + \frac{37}{9})$

$= (\frac{1\times5}{4\times5} - \frac{1\times4}{5\times4}) \times (\frac{7\times9}{1\times9} + \frac{37}{9})$

$= (\frac{5}{20} - \frac{4}{20}) \times (\frac{63}{9} + \frac{37}{9})$

$= \frac{1}{20} \times \frac{100}{9}$

$= \frac{100}{180}$

$= \frac{\cancel{2}\times\cancel{2}\times\cancel{5}\times5}{\cancel{2}\times\cancel{2}\times3\times3\times\cancel{5}}$

$= \frac{5}{9}$

Exercice n°2

a. $\frac{7}{3} \times \frac{5}{4} + \frac{1}{6}$

$= \frac{35}{12} + \frac{1 \times 2}{6 \times 2}$

$= \frac{35}{12} + \frac{2}{12}$

$= \frac{37}{12}$

b. $\frac{9}{28} \times \frac{7}{5} + \frac{10}{3} \times \frac{6}{25}$

$= \frac{9}{20} + \frac{4 \times 4}{5 \times 4}$

$= \frac{9}{20} + \frac{16}{20}$

$= \frac{25}{20}$

$= \frac{\cancel{5} \times 5}{2 \times 2 \times \cancel{5}}$

$= \frac{5}{4}$

Exercice n°3

a. $\frac{\frac{4}{3} - \frac{5}{6}}{\frac{1}{2} + 1}$

$= \frac{\frac{4 \times 2}{3 \times 2} - \frac{5}{6}}{\frac{1}{2} + \frac{1 \times 2}{1 \times 2}}$

$= \frac{\frac{8}{6} - \frac{5}{6}}{\frac{1}{2} + \frac{2}{2}}$

$$= \frac{\frac{3}{6}}{\frac{3}{2}}$$

$$= \frac{3}{6} \times \frac{2}{3}$$

$$= \frac{\cancel{3} \times \cancel{2}}{\cancel{2} \times \cancel{3} \times 3}$$

$$= \frac{1}{3}$$

b. $$\frac{2 - \frac{5}{12}}{\frac{1}{3} - \frac{8}{5}}$$

$$= \frac{\frac{2 \times 12}{1 \times 12} - \frac{5}{12}}{\frac{1 \times 5}{3 \times 5} - \frac{8 \times 3}{5 \times 3}}$$

$$= \frac{\frac{24}{12} - \frac{5}{12}}{\frac{5}{15} - \frac{24}{15}}$$

$$= \frac{\frac{19}{12}}{\frac{-19}{15}}$$

$$= \frac{19}{12} \times \frac{-15}{19}$$

$$= \frac{\cancel{19} \times (\cancel{-3}) \times 5}{2 \times 2 \times \cancel{3} \times \cancel{19}}$$

$$= \frac{-15}{12}$$

$$= \frac{\cancel{-3} \times 5}{2 \times 2 \times \cancel{3}}$$

$= \frac{-5}{4}$

Exercice n°4

a. $\frac{\frac{4}{3} - 1}{\frac{7}{6} - 2}$

$= \frac{\frac{4}{3} - \frac{1\times3}{1\times3}}{\frac{7}{6} - \frac{2\times6}{1\times6}}$

$= \frac{\frac{4}{3} - \frac{3}{3}}{\frac{7}{6} - \frac{12}{6}}$

$= \frac{\frac{1}{3}}{\frac{-5}{6}}$

$= \frac{1}{3} \times \frac{-6}{5}$

$= \frac{-2\times\cancel{3}}{\cancel{3}\times5}$

$= \frac{-2}{5}$

b. $\frac{\frac{2}{3} + \frac{1}{2}}{\frac{17}{9} - \frac{1}{3}}$

$= \frac{\frac{2\times2}{3\times2} + \frac{1\times3}{2\times3}}{\frac{17}{9} - \frac{1\times3}{3\times3}}$

$$= \frac{\frac{4}{6} + \frac{3}{6}}{\frac{17}{9} - \frac{3}{9}}$$

$$= \frac{\frac{7}{6}}{\frac{14}{9}}$$

$$= \frac{7}{6} \times \frac{9}{14}$$

$$= \frac{\cancel{7} \times \cancel{3} \times 3}{2 \times \cancel{3} \times 2 \times \cancel{7}}$$

$$= \frac{3}{4}$$

Exercice n°5

$$\mathbf{C} = \frac{22}{33} - \frac{35}{18} \times \frac{15}{7}$$

$$= \frac{2 \times \cancel{11}}{3 \times \cancel{11}} - \frac{5 \times \cancel{7} \times \cancel{3} \times 5}{2 \times \cancel{3} \times 3 \times \cancel{7}}$$

$$= \frac{2}{3} - \frac{25}{6}$$

$$= \frac{2 \times 2}{3 \times 2} - \frac{25}{6}$$

$$= \frac{4}{6} - \frac{25}{6}$$

$$= \frac{-21}{6}$$

$$= \frac{-\cancel{3} \times 7}{2 \times \cancel{3}}$$

$$= \frac{-7}{2}$$

$$\mathbf{A} = \frac{1 + \frac{2}{3}}{1 - \frac{2}{3}}$$

$$= \frac{\frac{1\times3}{1\times3} + \frac{2}{3}}{\frac{1\times3}{1\times3} - \frac{2}{3}}$$

$$= \frac{\frac{3}{3} + \frac{2}{3}}{\frac{3}{3} - \frac{2}{3}}$$

$$= \frac{\frac{5}{3}}{\frac{1}{3}}$$

$$= \frac{5}{3} \times \frac{3}{1}$$

$$= \frac{5\times\cancel{3}}{\cancel{3}}$$

$$= 5$$

$$\mathbf{T} = -6 - \frac{\frac{5}{15}}{2}$$

$$= -6 - \frac{\cancel{5}}{3\times\cancel{5}} \times \frac{1}{2}$$

$$= -6 - \frac{1}{6}$$

$$= \frac{-6\times6}{1\times6} - \frac{1}{6}$$

$$= \frac{-36}{6} - \frac{1}{6}$$

$$= \frac{-37}{6}$$